

The International Phonetic Alphabet (IPA)

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LINGUIST 101 People and Their Language
Sections AA & AB
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Plan for Today

1 Housekeeping

2 CFGs Review

3 The IPA

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Assignments

- Assignment #3 (CFGs and the IPA) is due **next Monday (Sep 28)!**
- Any questions on **Assignment #3?**
- Any question on **this week's lectures?**

Midterm #1

- Midterm #1 is on **October 14 (Wed)**, in-class
- It will cover CFGs and the IPA

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Recursion and infinity

(1) G_1 :

a. $S \rightarrow a b T$

b. $T \rightarrow T$

c. $T \rightarrow c$

- Does G_1 have recursion? **Yes!**
- Does G_1 generate a finite or infinite language? **Finite**

Recursion and infinity

(2) G_2 :

a. $S \rightarrow T a a$

b. $T \rightarrow T c$

c. $T \rightarrow b$

- Does G_2 have recursion? **Yes!**
- Does G_2 generate a finite or infinite language? **Infinite**

Recursion and infinity

- You can have: (i) a grammar that is recursive and infinite; (ii) recursive and finite; (iii) not recursive and finite
- You **cannot** have a grammar that is non-recursive and infinite!

	Recursive	Not recursive
Finite	✓	✓
Infinite	✓	✗

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The International Phonetic Alphabet

[a_ɪ p^hiː e_ɪ]

- A common vocabulary to represent the sounds of all languages
- There are over **100 letters** in the IPA! Thankfully, we will only focus on the sounds of English
- How many sounds does English have? Is the English orthography a good way to represent its sounds?
 - It varies depending on your dialect!
 - Usually, we say that English has 24 consonants
 - In American English, there are 14-16 vowels

English Consonants

										voiceless		voiced				
	Bilabial		Labio-dental		Dental		Alveolar		Post-alveolar		Palatal		Velar		Glottal	
Plosive	p	b					t	d					k	g	ʔ	
Affricate									tʃ	dʒ						
Nasal stop		m						n						ŋ		
Fricative			f	v	θ	ð	s	z	ʃ	ʒ					h	
Approximant		w						ɹ				j		w		
Lateral approximant								l								

[w] is specifically a voiced **labial-velar** approximant, rather than a purely bilabial or velar sound.

- What symbol represents the voiced post-alveolar affricate?
- /dʒ/ as in /laɪdʒ/ ‘large’
- What symbol represents the voiceless velar plosive (=stop)?
- /k/ as in /kæt/ ‘cat’

Minimal change

- In the IPA, consonants are represented along **three dimensions**
 - **Place of articulation** (x-axis)
 - **Manner of articulation** (y-axis)
 - **Voicing**
- Two words are **minimally different** (or, they form a **minimal pair**) if they vary along **exactly one dimension**
 - 1 /kæt/ 'cat' [voiceless, ~~velar~~ bilabial, plosive]
 - 2 → /pæt/ 'pat' [voiceless **voiced**, bilabial, plosive]
 - 3 → /bæt/ 'bat' [voiced, bilabial, ~~plosive~~ nasal]
 - 4 → /mæt/ 'mat' [voiced, bilabial, nasal]

Group activity

Let's work in groups now!

Have a great weekend!